Image Enhancement: A Lightweight, Easy-to-Use, and Efficient C++ Library

Algorithm Engineering 2026 Project

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Abstract

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Keywords

task parallelism, WebAssembly, OpenMP, image processing, C++

ACM Reference Format:

Daniel Motz, Leonard Teschner, and Mher Mnatsakanyan. 2026. Image Enhancement: A Lightweight, Easy-to-Use, and Efficient C++ Library: Algorithm Engineering 2026 Project. In *Proceedings of AEPRO 2026: Algorithm Engineering Projects (AEPRO 2026).* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

1.1 Background

The digitization of documents, especially images scanned or captured with smartphone cameras, is an important step in making information available [1, 15]. It also forms the basis for further processing steps such as optical character recognition (OCR). Existing tools and libraries for image enhancement and document preprocessing are powerful, but often too complex, too specialized, or have many dependencies. This makes it difficult to integrate them into small, user-defined programs and leads to the bloating of these programs. There is therefore a need for a lightweight, easy-to-use, and efficient solution that focuses on image enhancement while being flexible and modular enough to be used in a variety of applications. As modern smartphone cameras capture images with ever-higher resolutions, the computing time required to process them also increases. Efficient implementation that utilizes the advantages of parallelization and other optimization techniques is therefore important for improving processing speed and ensuring user satisfaction.

1.2 Related Work

Image enhancement and document preprocessing are covered by a wide range of open-source tools and commercial software. The scope ranges from general purpose image processing libraries and OCR (optical character recognition) libraries to specialized tools for post-processing scanned documents.

One of the most widely used libraries for general image processing is OpenCV [8]. Described by Bradski in 2000 [4], OpenCV offers a wide range of functions and algorithms for image processing, such as filters, transformations, segmentation, and feature detection, as well as machine and deep learning. OpenCV is written in C++ and has a modular, cross-platform architecture that allows for easy integration into various applications. However, OpenCV is a very complex library with many functions and dependencies, which leads to overhead when only a fraction of the functions are used. OpenCV is also licensed under the BSD license. Similar to OpenCV, ImageMagick [6] is a comprehensive suite for image processing that is available as a command line tool and as a library. It offers functions such as format conversion, resizing, filtering, and color correction. However, ImageMagick is not specifically designed to enhance scanned or photographed images, but rather for general image processing. The wide range of functions and possibilities results in a large number of dependencies and a large code base. ImageMagick is licensed under the Apache License.

The open source OCR engine Tesseract is often used for document analysis. Tesseract was described by Smith in 2007 [12] and has become one of the leading OCR engines. In addition to text recognition, Tesseract also offers some image enhancement features, but the focus is on OCR. The image enhancement features are typically integrated into the OCR workflow and are not available as general image enhancement features. Tesseract is licensed under the Apache License. A similar library specializing in OCR is the C-based Leptonica library [3]. Leptonica offers image processing functions that are widely used in the backend of OCR engines such as Tesseract. It provides functions such as binarization, noise reduction, and morphological operations, which are important for preparing images for OCR recognition. Although Leptonica is relatively lightweight compared to the other libraries, it primarily serves as a low-level library and does not offer ready-to-use, configurable processing pipelines, for example. It is also written in C, which makes integration into C++ applications difficult. Leptonica is licensed under the BSD license.

A more specialized tool is Unpaper [10], which focuses on post-processing scanned documents. It offers features such as deskewing, binarization, and noise reduction that have been developed specifically for this purpose. However, Unpaper is highly specialized and cannot be flexibly integrated into custom C++ applications. Due to its configuration, Unpaper has many dependencies, which makes it difficult to use. Unpaper is licensed under the GPL license, which restricts commercial use.

1.3 Our Contribution

Unlike the existing tools and libraries described, our approach is designed to be minimalistic, extensible, and user-friendly. Instead of being a comprehensive image processing library or OCR engine, we focus on a lightweight, modular image enhancement pipeline

written entirely in C++ with only one dependency (CImg [13]), which is itself lightweight and portable. We systematically optimize and improve our intuitive baseline implementation and evaluate the performance gains through various experiments. Compared to OpenCV and ImageMagick, our work is characterized by its deliberate focus on image enhancement and compact architecture. Unlike Tesseract and Leptonica, our pipeline is not integrated into an OCR workflow and limited by low-level primitive functions, but offers a configurable standalone solution that is ready for immediate use. Our pipeline is not exclusively specialized in the post-processing of scanned documents, but is designed as a general image enhancement pipeline that can be used for a variety of applications. In addition, our licensing model (MIT-0 or CeCILL-C) ensures its use in research and commercial applications. By combining minimal dependencies, a modular architecture, portability through C++ and OpenMP, and a commercially usable license, our pipeline fills a gap between large-scale, general-purpose image processing libraries and highly specialized tools for document enhancement.

1.4 Outline

In section 2, we introduce the pipeline that we will implement and optimize. In section 3, we describe the optimization techniques used. These are evaluated in section 4, where we compare the performance of the pipeline under different settings to the baseline implementation. Finally, we summarize the results in section 5 and provide an outlook on possible future work.

2 Pipeline

The pipeline is designed to be flexible and modular, enabling users to activate, deactivate and configure each step as required. Each step is designed to run independently of the others, except for the conversion to grayscale step 1, which forms the basis for subsequent operations. The steps in the pipeline are as follows:

(1) Grayscale Conversion: Conversion of the input image into a grayscale image to simplify processing and reduce computing time.
(2) Deskewing: Correction of skewed images to improve readability and OCR accuracy.
(3) Contrast Enhancement: Improvement of the contrast in the image to facilitate the distinction between text and background.
(4) Denoising: Removal of noise and artifacts in the image to improve clarity.
(5) Binarization: Segmentation of the grayscale image into a binary image consisting of foreground and background.
(6) Morphological Operations: Application of operations such as erosion and dilation to improve the structure of the text and remove small errors.
(7) Color preservation: The color information of the original image is preserved at the locations of the foreground (especially relevant for historical documents to preserve their original coloration).

Several algorithms are implemented in the steps 4, 5 and 6. As these have different effects on the result, users can select the algorithm best suited to their application. For instance, Bataineh's adaptive binarisation method with automatic window adjustment [2] is helpful when binarising historical, handwritten documents with uneven writing and lighting, whereas Otsu's method [11] is a good choice for binarising typewritten documents with uniform font and lighting. Standard configurations are provided for each step, enabling the pipeline to be used straight away. These configurations deliver good results for most use cases and are based on recommendations from the authors. This means that the pipeline can be used immediately, without the need for additional information. However, users can also customise the configurations to optimise the pipeline and improve their specific results. The pipeline methods 1 - 7 can be used individually by users, as a library in their own program for their specific use case.

We provide a command-line application that can be used independently of a program to access the pipeline. This enables the pipeline to be configured and used directly via the command line.

3 Pipeline Optimizations

3.1 Parallelization and Concurrency

To achieve better performance on multicore processors, the individual steps are parallelized using OpenMP. *parallel for collapse(3)* environments are mainly used for the pixel processing loops. However, some binarization algorithms also use other parallelization strategies such as the reduction clauses *reduction(min)* and *recuction(max)*. When implementing parallelization, care is taken to ensure that no race conditions or false sharing occur. For example, thread-local buffers are used for contrast enhancement to avoid false sharing and locking. In the deskewing logic, detection and application of the correction are separated. The angle is detected once and then applied to the grayscale and color image to avoid unnecessary calculations. Intuitively, this was done twice.

3.2 Memory Management and Allocation

To prevent increased memory consumption and long runtimes, the image is not copied at each step. Instead, the steps work in-place on the input image. For this purpose, a pointer is passed at each step. This approach is continued in grayscale conversion by replacing the final deep copy task of the image with a pointer swap of the original image pointer with the new one. Memory optimizations were made in further steps. For example, in the adaptive Gaussian blur filter and the adaptive median filter, the buffer allocations are removed from the pixel processing loop to reduce the number of memory allocations. Sauvola's binarization algorithm uses a lightweight fused vector builder instead of the *8-bit CImg<double>* image structure. Similarly, the core utilities return preallocated buffers instead of heavy objects.

3.3 Algorithmic Complexity Reduction

Several different algorithms are implemented in the pipeline. Some of these are optimized through algorithmic improvements to reduce complexity. One example is Bataineh binarization, which reduces the number of operations per pixel from $O(W^2)$ to $O(1)$ by using integral images to calculate the number of black and red pixels in the window instead of iterating over the entire window. In morphological operations, the complexity is reduced from $O(K^2)$ to $O(2K)$

by decomposing the 2D kernel operations into two 1D kernel operations for the horizontal and vertical dimensions. In addition, the sliding window min/max algorithm is implemented with a monotonic queue to enable computation for arbitrary window sizes in amortized $O(1)$ time. In the deskewing logic, the "rotate and project" method is replaced by a "Radon transform" method. Instead of moving the pixels, the coordinates of the foreground are extracted and mathematically projected.

3.4 Loop level Optimizations

To reduce the runtime of pixel processing loops, logical optimizations are made to the main loops that iterate over the pixels. In contrast enhancement, for example, the linear stretching formula is calculated in advance and stored in a lookup table. As a result, the main loop performs an $O(1)$ query per loop iteration instead of floating-point calculations. In addition, the if/else branch in the pixel processing loop is replaced by std::min clamping. This allows the compiler to auto-vectorize the loop with SIMD. In Sauvola binarization, a branchless lookup is also implemented to improve performance.

3.5 Architectial Refactoring

In some cases, architectural changes were also made to improve performance. For example, in the deskewing logic, detection and application of the correction were separated. The angle is first detected and then applied to the grayscale and color images to avoid unnecessary calculations. In the core utilities, pre-built buffers are returned instead of creating heavy objects. This improves performance.

4 Experiments

4.1 Setup

To evaluate the performance of the pipeline with the optimizations, various experiments are conducted and the results are compared with the times of the intuitive implementation. The experiments examine the runtime when processing images of different sizes with and without parallelization. The experiments are conducted on a computer with an Intel Core i5-14600kf processor with six performance cores and eight efficiency cores. The computer has 32 GB of RAM. They are run on Windows 11 with the Windows Subsystem for Linux (WSL). Care is taken to ensure that no background applications are running and that power-saving mode is disabled so as not to affect performance. The fans are also activated to prevent temperature throttling.

Test images are individual images from different OCR datasets. One test image is document text from the FUNSD dataset [5], which has been scanned and contains a lot of noise. Two additional test images are handwritten documents from the HTR ICFHR 2016 dataset [14]. Both of these images have a resolution of XxY and exhibit bleed-through, whereby text from other pages is visible. For colour documents, a test image from the LRDE Document Binarisation Dataset [9] is used. This is a scanned excerpt from a French magazine of very good quality containing many small sections of text. It has a resolution of XxY. A test image from the KITA imaging programme Forum is used to demonstrate a combination of historical text, UI elements and colour. All images are provided as JPG files.

To test performance with larger images, the KITA image, which has an initial resolution of XxY, is upscaled to 2k, 4k, 8k and 16k using the ImageUpscaler tool [7]. All images are processed using the pipeline and the runtime is measured for each step. Additionally, the images are processed using one and eight threads to evaluate the effects of parallelisation.

4.2 Results

Figures **??** and **??** show the development of runtime with increasing image resolution for the intuitive and optimized implementations. It can be seen that runtime increases significantly with increasing image size, especially in the intuitive implementation. With the optimized implementation, the runtime is significantly lower, especially for larger images. At 16K resolution, the runtime of the intuitive implementation could no longer be calculated because small rounding errors in the deskewing step meant that the recognized angles in the color image and the grayscale image were no longer the same. As a result, the images could no longer be merged and the image could not be calculated. This problem is solved by the optimizations in the deskewing step, as the detection and application of the correction are separated here.

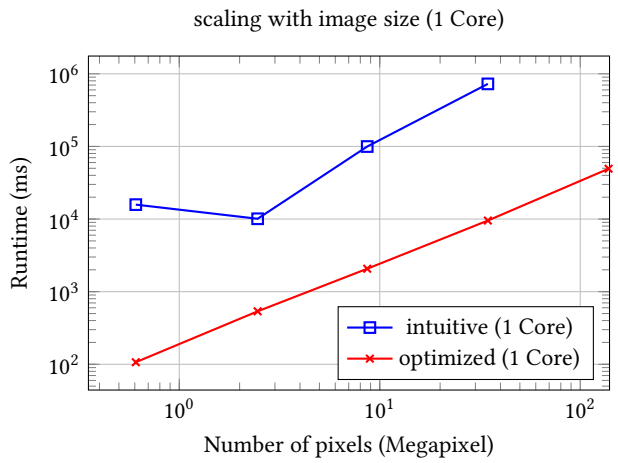

Figure 1: Runtime development with increasing image resolution.

The results of the individual phases show a significant reduction in runtime in the deskewing and Bataineh binarization phases when comparing the intuitive and optimized implementations. Table 2 shows the runtime of the individual phases for the KITA test image with a resolution of 3840x2254 on 8 cores. It can be seen that the optimizations lead to a massive reduction in phase runtime of over 98% for Bataineh binarization in particular. The deskewing phase also benefits from the optimizations and shows a reduction in phase runtime of over 98%. In other phases, such as grayscale and the filter, a significant reduction in phase runtime can also be observed, while in the morphological ops phase, a slight increase in phase runtime can be seen. WHY?????. Overall, this leads to a reduction in the total runtime of over 95%. In absolute terms, the runtime improvement is mainly due to the optimization of Bataineh binarization, as this

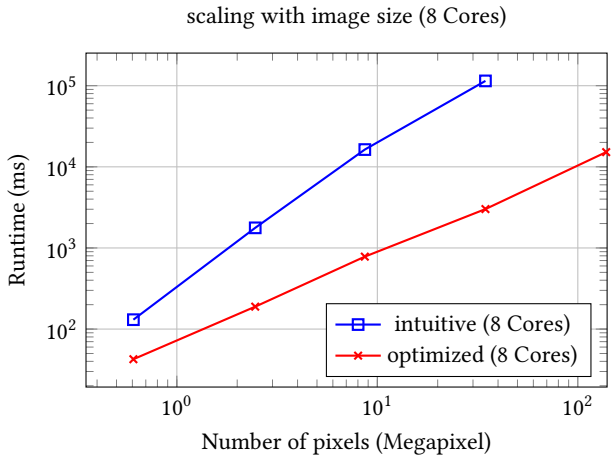

Figure 2: Runtime development with increasing image resolution.

phase accounts for the largest share of the total runtime in the intuitive implementation.

Table 1: Comparison of total runtimes (8 Cores)

Testbild	resolution	intuitive (ms)	optimized (ms)	Speedup
Text Small 1K	1017x597	130,577	42,511	3,07×
Text 2K	2048x1202	1.770,138	188,975	9,37×
Magazine 2K	2516x3272	16.701,170	649,924	25,70×
Text 4K	3840x2254	16.328,084	781,634	20,89×
Text 8K	7680x4508	114.607,067	3.015,428	38,01×
Text 16K	15360x9017	-*	15.181,484	-*

*Could not be calculated due to massive runtime

Table 2: Comparison of phase runtimes (3840x2254, 8 Cores)

Phase	intuitive (ms)	optimized (ms)	diff (%)
Init & Copy	65,458	63,672	-2,728%
Grayscale	30,950	17,753	-42,640%
Deskew	**159,405**	**3,062**	**-98,079%**
Filters (Sum)	483,312	380,467	-21,279%
Bataineh	**15.507,105**	**217,347**	**-98,598%**
Morphological Ops	66,663	93,05	+39,583%
Color Preservation	4,129	4,089	-0,969%
Total	**16.328,084**	**781,634**	**-95,213%**

Table 3 shows the detailed phase runtime for the different phases and images. It can be seen that the optimizations lead to a significant reduction in runtime in all phases, especially in Bataineh binarization, which accounts for the largest share of the total runtime in the intuitive implementation. For larger images such as the 8K text image, it can be seen that the image becomes too large for the system, as there is a sharp increase in the Init & Copy phase, which indicates a memory limitation.

5 Conclusions

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Table 3: Detaillierte Laufzeit der Algorithmus-Phasen in Millisekunden (Auszug)

Phase	1K Small	2K Text	2K Mag.	4K Text	8K Text
Init & Copy	1,29	4,20	61,83	65,46	241,93
Grayscale	0,60	2,95	27,68	30,95	92,45
Deskew	24,22	59,89	28,90	159,41	518,54
Binarization	71,00	1.555,57	15.999,83	15.507,11	111.495,99
Filter (Sum)	28,74	125,14	520,73	483,31	1.968,43
TOTAL	**130,58**	**1.770,14**	**16.701,17**	**16.328,08**	**114.607,07**